COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS

RECURRENT SELF-ORGANISING NETWORKS

Master thesis

2022 Bc. Ema Adamová

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Department:	Department of Applied Informatics
Supervisor:	prof. RNDr. Kristína Malinovská, PhD.
Consultant:	

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Bc. Ema Adamová

35859437

Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

PRIHLÁŠKA NA ZÁVEREČNÚ PRÁCU

Meno a priezvisko študenta: Bc. Ema Adamová
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Anotácia: Self-organizing maps (SOMs) [1] are one of the classical neural network models that do not require a training signal and yet capture similarities in the training data and do so in a topological manner, that is, they capture similarities between data as distances in space. There are also recurrent versions such as MSOM [2] or RecSOM that are capable of capturing sequences of data in this way. The MSOMs are utilized, for example, in cognitive robotics. In classical SOMs, the values of the weights of the neurons in the map can be considered as models of the data that the network has learned. However, for recurrent SOMs, this relationship is not straightforward and reconstructing learned sequences is an interesting problem [2]. Another interesting phenomenon that has been addressed using self-organization is continuous learning and avoiding catastrophic forgetting [3]. The implementation and testing of the aforementioned algorithms related to recurrent self-organizing networks is very important for their use in future research.

Cieľ: The goal of the thesis is to implement recurrent SOM, namely MSOM and RecSOM in Python and test the implementation on suitable known test problems. Furthermore, the thesis shall investigate the possibility of reconstructing the stored sequences as well as the possibility of visualizing the information in the network and to implement such algorithms. The next task is to study and implement continuous learning for sequence self-organization and test the solution in a catastrophic interference problem. The final goal of this thesis is to integrate and test the proposed implementation in the context of ongoing research in cognitive robotics.

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[1] Kohonen, T., 1990. The self-organizing map. Proceedings of the IEEE, 78(9), pp.1464-1480.
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Vedúci: RNDr. Kristína Malinovská, PhD.
Katedra: FMFI.KAI - Katedra aplikovanej informatiky

35859437

Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

Vedúci katedry: doc. RNDr. Tatiana Jajcayová, PhD.

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